THE CATHOLIC UNIVERSITY OF EASTERN AFRICA

FACULTY OF SCIENCE

DEPARTMENT OF COMPUTER SCIENCE

UNIT: CMT 429

UNIT TITLE: INTRODUCTION TO DATA SCIENCE

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Phase 3 Project

- Domain: Business (Bank Marketing)
- Dataset: Bank Marketing Dataset (UCI Machine Learning Repository)
- Github: DS_Avengers_Bank_Marketing

Project Overview

This project focuses on using classification and regression techniques to analyze the Bank Marketing dataset from a Portuguese banking institution. The dataset captures results of direct marketing campaigns (via phone calls) where the main goal was to determine whether a client would subscribe to a term deposit. Using the R programming language, we aim to build predictive models that will support decision-making in future marketing campaigns.

Business Understanding

Business Problem

Banks spend significant resources on marketing campaigns, but most calls end without success. This leads to wasted time, money, and reduced customer trust. The main problem is how to predict which clients are more likely to subscribe to a term deposit, so that campaigns can be more targeted and cost-effective.

Stakeholders

- Bank management: Needs insights to improve campaign efficiency.
- Marketing teams: Want to know which customers to prioritize.
- Customers: Benefit from receiving more relevant and personalized offers.

Research Questions

1. What customer demographic and financial characteristics influence subscription decisions?
2. How do campaign-related variables affect customer response?
3. Can we build a model that accurately predicts whether a client will subscribe to a term deposit?
4. Which features are the most important for prediction?

Justification

The need for predictive modeling in banking is crucial because it reduces unnecessary calls, improves customer experience, and increases return on investment in marketing.

Data science provides tools to analyze historical data and develop models that support smarter decision-making

Data Description

- Source: UCI Machine Learning Repository – Bank Marketing Dataset.
- Instances: 45,211 records (depending on dataset version used).
- Key Features: Age, job, marital status, education, default, balance, housing loan, personal loan, contact type, campaign details, and previous campaign outcomes.
- Target Variable: y (binary: “yes” = client subscribed, “no” = client did not subscribe).
- Data Type: Mix of categorical, integer, and binary features.
- Missing Values: Minimal, mostly in categorical variables such as “pdays” and “poutcome”.

Objectives

Main Objective To build a predictive classification model that determines whether a client will subscribe to a term deposit.

Specific Objectives

1. Perform data preprocessing (cleaning, handling missing values, encoding).
2. Conduct exploratory data analysis (EDA) to identify trends and patterns.
3. Build and train classification models (logistic regression, decision tree, random forest).
4. Evaluate performance using accuracy, precision, recall, and F1-score.
5. Apply cross-validation for robustness.
6. Provide actionable recommendations for banking decision-makers.

Exploratory Data Analysis (EDA)

Univariate Analysis (Expected):

- Distribution of age, balance, campaign contacts.
- Frequency of categorical variables such as job type, marital status, and education.

Bivariate Analysis (Expected):

- Relationship between y (subscription) and independent features.
- Example: Subscription rate vs. education level, or vs. contact type.

Multivariate Analysis (Expected):

- Interaction between multiple features such as job, education, and balance in predicting y.
- Use of correlation plots and dimensionality reduction techniques.

Models Tested

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost
- Support Vector Machine

Dataset Selection Justification

In this project, we used the `bank-additional-full.csv` dataset and chose not to use `bank.csv`, `bank-full.csv`, or `bank-additional.csv`. The reason is that both `bank.csv` and `bank-additional.csv` are only 10% of the full data taken from `bank-full.csv` and `bank-additional-full.csv`, respectively.

We selected `bank-additional-full.csv` because it is a more complete and updated version of the original dataset. It includes five extra features related to the economy, such as employment rate and consumer confidence. These additional details help us better understand how external factors may affect customer decisions.

Using the full dataset gives us more accurate results and stronger insights for building predictive models.

In [2]:
```
# install.packages("dplyr")
library(dplyr) # Data wrangling
# install.packages("purrr")
library(purrr)
# install.packages("ggplot2")
library(ggplot2)     # Data visualization
# install.packages("htmlwidgets")
# install.packages("plotly")
library(plotly)
# install.packages("ggmosaic")
library(ggmosaic)
# install.packages("psych")
library(psych)
# install.packages("corrplot")
library(corrplot) # Correlation visualization
# install.packages("vcd")
library(vcd)  # for assocstats (Cramér's V)
#install.packages("caret")
library(caret) # encoding
# install.packages("randomForest")
library(randomForest)
# install.packages("rpart")
library(rpart)
# install.packages("rpart.plot")
library(rpart.plot)
# install.packages("xgboost")
library(xgboost)
# install.packages("e1071")
library(e1071)
```

In [3]:
```
# Load the dataset
data <- read.csv("data/bank-additional-full.csv", header = TRUE)
```

In [4]:
```
# Check dimensions (rows, columns)
dim(data)
```

41188 · 21

Dimensions: 21 columns (Features) and 41,188 rows.

In [5]:
```
# View the first rows
head(data)
```

	age	job	marital	education	default	housing	loan	contact	month
	<int>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
1	56	housemaid	married	basic.4y	no	no	no	telephone	may
2	57	services	married	high.school	unknown	no	no	telephone	may
3	37	services	married	high.school	no	yes	no	telephone	may
4	40	admin.	married	basic.6y	no	no	no	telephone	may
5	56	services	married	high.school	no	no	yes	telephone	may
6	45	services	married	basic.9y	unknown	no	no	telephone	may

In [6]:
```
# Summarizing Data
summary(data)
```

```
     age            job              marital           education
Min.   :17.00   Length:41188       Length:41188       Length:41188
1st Qu.:32.00   Class :character   Class :character   Class :character
Median :38.00   Mode  :character   Mode  :character   Mode  :character
Mean   :40.02
3rd Qu.:47.00
Max.   :98.00
  default            housing              loan             contact
Length:41188       Length:41188       Length:41188       Length:41188
Class :character   Class :character   Class :character   Class :character
Mode  :character   Mode  :character   Mode  :character   Mode  :character



   month           day_of_week           duration         campaign
Length:41188       Length:41188       Min.   :   0.0   Min.   : 1.000
Class :character   Class :character   1st Qu.: 102.0   1st Qu.: 1.000
Mode  :character   Mode  :character   Median : 180.0   Median : 2.000
                                      Mean   : 258.3   Mean   : 2.568
                                      3rd Qu.: 319.0   3rd Qu.: 3.000
                                      Max.   :4918.0   Max.   :56.000
    pdays          previous       poutcome          emp.var.rate
Min.   :  0.0   Min.   :0.000   Length:41188       Min.   :-3.40000
1st Qu.:999.0   1st Qu.:0.000   Class :character   1st Qu.:-1.80000
Median :999.0   Median :0.000   Mode  :character   Median : 1.10000
Mean   :962.5   Mean   :0.173                      Mean   : 0.08189
3rd Qu.:999.0   3rd Qu.:0.000                      3rd Qu.: 1.40000
Max.   :999.0   Max.   :7.000                      Max.   : 1.40000
cons.price.idx  cons.conf.idx     euribor3m      nr.employed
Min.   :92.20   Min.   :-50.8   Min.   :0.634   Min.   :4964
1st Qu.:93.08   1st Qu.:-42.7   1st Qu.:1.344   1st Qu.:5099
Median :93.75   Median :-41.8   Median :4.857   Median :5191
Mean   :93.58   Mean   :-40.5   Mean   :3.621   Mean   :5167
3rd Qu.:93.99   3rd Qu.:-36.4   3rd Qu.:4.961   3rd Qu.:5228
Max.   :94.77   Max.   :-26.9   Max.   :5.045   Max.   :5228
     y
Length:41188
Class :character
Mode  :character
```

Dataset Overview

- Size: 41,188 observations
- Structure: Mix of numeric and categorical variables

Demographics

- Age: Ranges from 17 to 98 years; average age is 40
- Job, Marital Status, Education: All are categorical with varied entries

Financial Attributes

- Default, Housing, Loan: Categorical indicators of financial status
- Contact Method: Categorical (e.g., phone, cellular)

Campaign Details

- Month & Day of Week: Categorical indicators of contact timing
- Duration: Call duration ranges from 0 to 4918 seconds median is 180 seconds
- Campaign: Number of contacts per client ranges from 1 to 56

Historical & Economic Indicators

- Pdays: Mostly 999 (indicating no prior contact)
- Previous Contacts: Mostly 0, max is 7
- Poutcome: Outcome of previous marketing campaigns (categorical)
- Economic Variables:
  - Employment Variation Rate: Avg ~0.08
  - Consumer Price Index: Avg ~93.58
  - Consumer Confidence Index: Avg ~-40.5
  - Euribor 3-month Rate: Avg ~3.62
  - Number of Employees: Avg ~5167

Target Variable

- y: Indicates if the client subscribed to a term deposit (binary categorical)

In [7]:
```
# View data set structure and data types
str(data)
```

```
'data.frame':	41188 obs. of  21 variables:
 $ age           : int  56 57 37 40 56 45 59 41 24 25 ...
 $ job           : chr  "housemaid" "services" "services" "admin." ...
 $ marital       : chr  "married" "married" "married" "married" ...
 $ education     : chr  "basic.4y" "high.school" "high.school" "basic.6y" ...
 $ default       : chr  "no" "unknown" "no" "no" ...
 $ housing       : chr  "no" "no" "yes" "no" ...
 $ loan          : chr  "no" "no" "no" "no" ...
 $ contact       : chr  "telephone" "telephone" "telephone" "telephone" ...
 $ month         : chr  "may" "may" "may" "may" ...
 $ day_of_week   : chr  "mon" "mon" "mon" "mon" ...
 $ duration      : int  261 149 226 151 307 198 139 217 380 50 ...
 $ campaign      : int  1 1 1 1 1 1 1 1 1 1 ...
 $ pdays         : int  999 999 999 999 999 999 999 999 999 999 ...
 $ previous      : int  0 0 0 0 0 0 0 0 0 0 ...
 $ poutcome      : chr  "nonexistent" "nonexistent" "nonexistent" "nonexistent"
...
 $ emp.var.rate  : num  1.1 1.1 1.1 1.1 1.1 1.1 1.1 1.1 1.1 1.1 ...
 $ cons.price.idx: num  94 94 94 94 94 ...
 $ cons.conf.idx : num  -36.4 -36.4 -36.4 -36.4 -36.4 -36.4 -36.4 -36.4 -36.4 -3
6.4 ...
 $ euribor3m     : num  4.86 4.86 4.86 4.86 4.86 ...
 $ nr.employed   : int  5191 5191 5191 5191 5191 5191 5191 5191 5191 5191 ...
 $ y             : chr  "no" "no" "no" "no" ...
```

Attribute Description

Bank Client Information: These features describe the personal and financial background of each client:

- `age` : Client's age (numeric)
- `job` : Type of job (e.g., admin, technician, student, retired, unknown)
- `marital` : Marital status (e.g., married, single, divorced/widowed, unknown)
- `education` : Education level (e.g., basic, high school, university, professional course, unknown)
- `default` : Whether the client has credit in default (yes, no, unknown)
- `housing` : Whether the client has a housing loan (yes, no, unknown)
- `loan` : Whether the client has a personal loan (yes, no, unknown)

Last Contact Details (Current Campaign): These features describe the last time the client was contacted during the current marketing campaign:

- `contact` : Communication type used (cellular or telephone)
- `month` : Month of last contact (e.g., jan, feb, mar, ..., dec)
- `day_of_week` : Day of the week of last contact (e.g., mon, tue, wed, thu, fri)
- `duration` : Duration of the last contact in seconds (numeric) Note: This feature strongly influences the outcome but is only known after the call. For realistic modeling, it shall be excluded.

Campaign History: These features track how often and when the client was contacted:

- `campaign` : Number of contacts during the current campaign (includes last contact)
- `pdays` : Days since the client was last contacted in a previous campaign (999 means never contacted)
- `previous` : Number of contacts before the current campaign
- `poutcome` : Outcome of the previous campaign (failure, success, nonexistent)

Social and Economic Indicators: These features reflect the broader economic environment at the time of contact:

- `emp.var.rate` : Employment variation rate (quarterly)
- `cons.price.idx` : Consumer price index (monthly)
- `cons.conf.idx` : Consumer confidence index (monthly)
- `euribor3m` : Euribor 3-month interest rate (daily)
- `nr.employed` : Number of employees in the economy (quarterly)

Target Variable

- `y` : Whether the client subscribed to a term deposit (yes or no)

Data Preprocessing

In [8]:
```
# Count target variable values
table(data$y)
```

```
   no   yes 
36548  4640 
```

Target Variable: 4640 clients subscribed to a term deposit while 36548 did not.

Remove Unnecessary Columns

- 'duration' since it's only known after the call

In [9]:
```
# Remove 'duration'
data$duration <- NULL
```

Rename Columns for clarity

In [10]:
```
# Rename columns
data <- data %>%
  rename(
    marital_status = marital,
    education_level = education,
    has_default = default,
    has_housing_loan = housing,
    has_personal_loan = loan,
    contact_type = contact,
    month_of_contact = month,
    day_of_week_contact = day_of_week,
    num_contacts = campaign,
    days_since_last_contact = pdays,
    num_prev_contacts = previous,
    prev_campaign_outcome = poutcome,
    employment_rate = emp.var.rate,
    consumer_price_index = cons.price.idx,
    consumer_confidence_index = cons.conf.idx,
    euribor_rate = euribor3m,
    num_employees = nr.employed
  )
```

In [11]:
```
colnames(data)
```

'age' · 'job' · 'marital_status' · 'education_level' · 'has_default' · 'has_housing_loan' · 'has_personal_loan' · 'contact_type' · 'month_of_contact' · 'day_of_week_contact' · 'num_contacts' · 'days_since_last_contact' · 'num_prev_contacts' · 'prev_campaign_outcome' · 'employment_rate' · 'consumer_price_index' · 'consumer_confidence_index' · 'euribor_rate' · 'num_employees' · 'y'

Handle Missing Values

In [12]:
```
# Count missing values per column
colSums(is.na(data))
```

age: 0 **job:** 0 **marital_status:** 0 **education_level:** 0 **has_default:** 0 **has_housing_loan:** 0 **has_personal_loan:** 0 **contact_type:** 0 **month_of_contact:** 0 **day_of_week_contact:** 0 **num_contacts:** 0 **days_since_last_contact:** 0 **num_prev_contacts:** 0 **prev_campaign_outcome:** 0 **employment_rate:** 0 **consumer_price_index:** 0 **consumer_confidence_index:** 0 **euribor_rate:** 0 **num_employees:** 0 **y:** 0

Missing Values: There are no visible missing values except some categorical features contain missing values labeled as `"unknown"`. These can be handled by:

- Treating `"unknown"` as a valid category

- Removing affected rows
- Using imputation techniques to fill in missing data

In [13]:
```
# get categorical columns
categorical_cols <- names(data)[sapply(data, function(x) is.factor(x) || is.char

categorical_cols
```

'job' · 'marital_status' · 'education_level' · 'has_default' · 'has_housing_loan' · 'has_personal_loan' · 'contact_type' · 'month_of_contact' · 'day_of_week_contact' · 'prev_campaign_outcome' · 'y'

In [14]:
```
# Check which columns contain "unknown"
sapply(data[categorical_cols], function(x){
    round(mean(x == "unknown") * 100, 2)
})
```

job: 0.8 **marital_status:** 0.19 **education_level:** 4.2 **has_default:** 20.87 **has_housing_loan:** 2.4 **has_personal_loan:** 2.4 **contact_type:** 0 **month_of_contact:** 0 **day_of_week_contact:** 0 **prev_campaign_outcome:** 0 **y:** 0

In [15]:
```
# remove rows where low "unknown"

unknown_cols <- c("job", "marital_status", "education_level", "has_housing_loan"

for(col in unknown_cols){
  data <- data[data[[col]] != "unknown", ]
}
```

Remove Outliers (Using IQR Method on Numeric Variables)

In [16]:
```
# Filter numeric columns
numeric_cols <- names(data)[sapply(data, is.numeric)]

# Function to filter outliers
remove_outliers <- function(df, col) {
  Q1 <- quantile(df[[col]], 0.25, na.rm = TRUE)
  Q3 <- quantile(df[[col]], 0.75, na.rm = TRUE)
  IQR <- Q3 - Q1
  lower <- Q1 - 1.5 * IQR
  upper <- Q3 + 1.5 * IQR
  df %>% filter(.data[[col]] >= lower & .data[[col]] <= upper)
}

# Remove outliers
data <- reduce(numeric_cols, remove_outliers, .init = data)
```

In [17]:
```
dim(data)
```

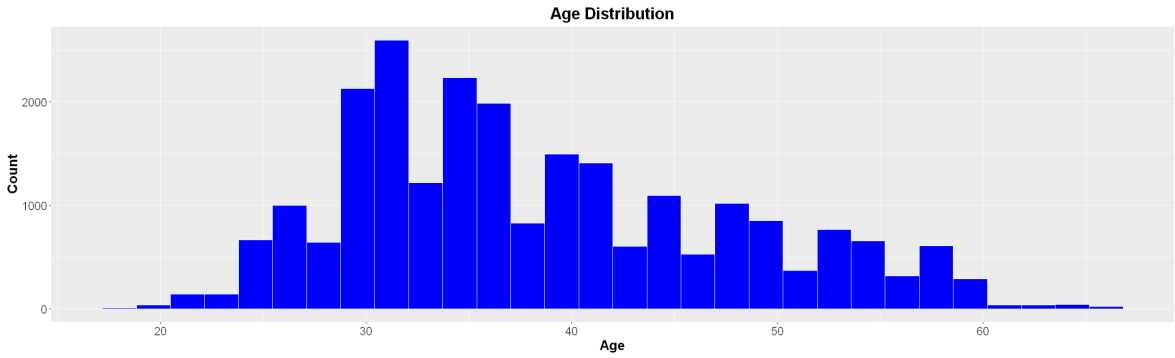
23783 · 20

EDA (Exploratory Data Analysis)

Univariate Analysis

Age Distribution

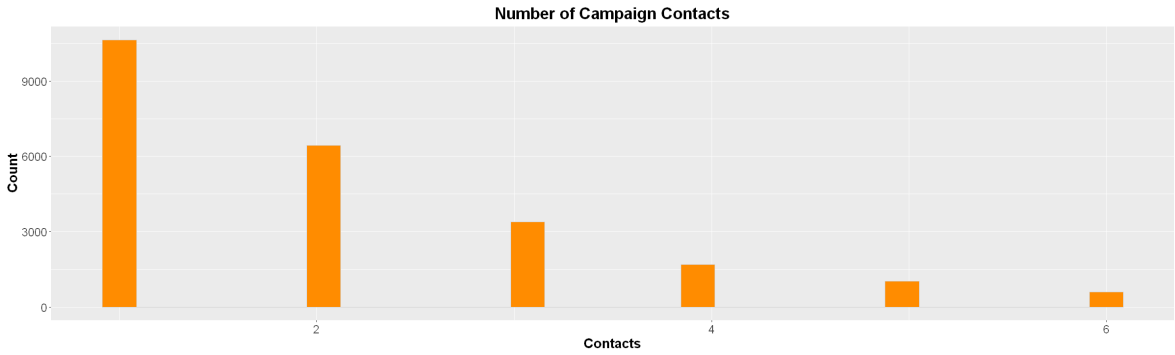
In [18]:
```
# Width and Height
options(repr.plot.width = 20, repr.plot.height = 6)

ggplot(data, aes(x = age)) +
  geom_histogram(bins = 30, fill = "blue", color = "white") +
  labs(title = "Age Distribution", x = "Age", y = "Count") +
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14),
    axis.text.y  = element_text(size = 14)
  )
```



- Most clients are aged 30–35, indicating strong engagement from younger working adults.
- The distribution is right-skewed, with fewer older clients (50+), suggesting lower outreach or interest in that segment.
- Under-20s are rare, highlighting a gap in youth targeting.
- These trends shows marketing segmentation and product design needs tailoring offers to high-response age groups and exploring strategies for underserved ones.

Distribution of Campaign Contacts (Numeric)

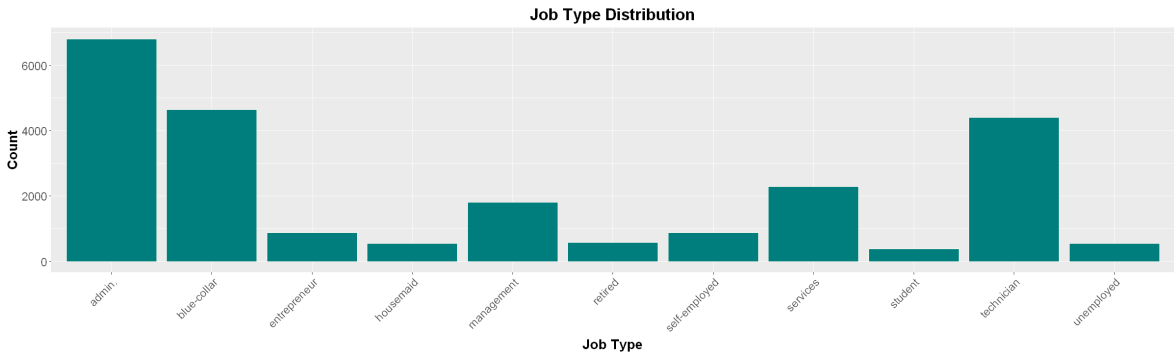
In [19]:
```
# Width and Height
options(repr.plot.width = 20, repr.plot.height = 6)

ggplot(data, aes(x = num_contacts)) +
  geom_histogram(bins = 30, fill = "darkorange", color = "lightgray") +
  labs(title = "Number of Campaign Contacts", x = "Contacts", y = "Count") +
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14),
    axis.text.y  = element_text(size = 14)
  )
```



- Most clients were contacted only once, indicating a single-touch strategy dominates.
- Contact frequency drops sharply after the first attempt, suggesting limited follow-up.
- Low multi-contact engagement may affect conversion rates, repeated contact could improve outcomes.
- Distribution is right-skewed, showing diminishing returns or reduced effort beyond initial outreach.

Job Type Frequency

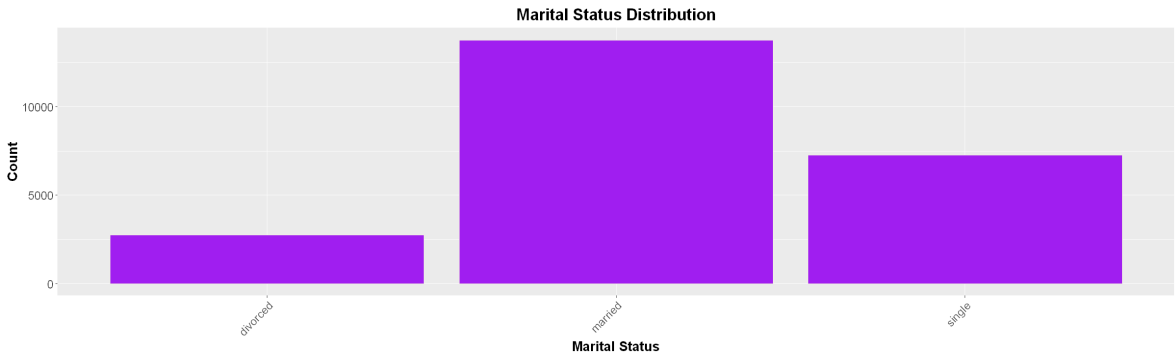
In [20]:
```
options(repr.plot.width = 20, repr.plot.height = 6)
# color teal and ivory
ggplot(data, aes(x = job)) +
  geom_bar(fill = "#008080", color="#FFFFF0") +
  labs(title = "Job Type Distribution", x = "Job Type", y = "Count") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14),
    axis.text.y  = element_text(size = 14)
  )
```



- Blue-collar and admin roles dominate, indicating strong representation from manual labor and leadership segments.
- Technicians and service workers also form a significant portion of the client base.
- Students, housemaids, and unknown job types are underrepresented, suggesting limited outreach or engagement in these groups.
- Diverse job categories imply the campaign reaches a broad employment spectrum, useful for tailoring offers by profession.

Marital Status Frequency

In [21]:
```
options(repr.plot.width = 20, repr.plot.height = 6)

ggplot(data, aes(x = marital_status)) +
  geom_bar(fill = "purple", color="beige") +
  labs(title = "Marital Status Distribution", x = "Marital Status", y = "Count")
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14),
    axis.text.y  = element_text(size = 14)
  )
```

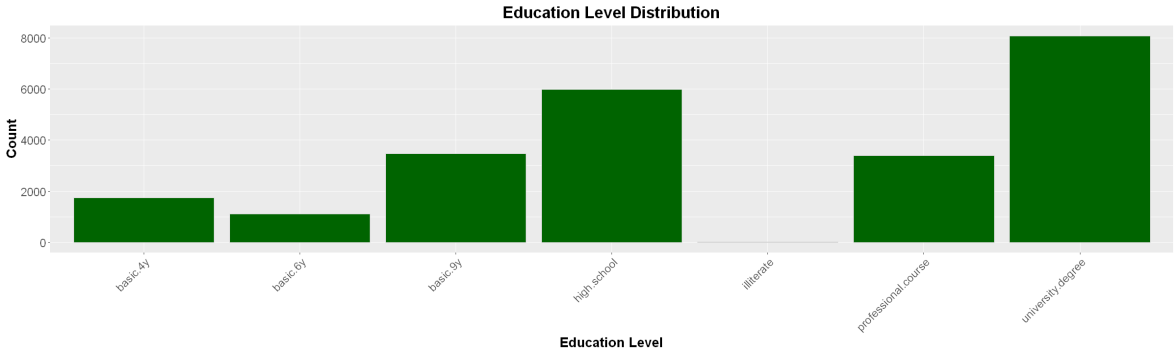


- Married clients dominate, indicating strong engagement from this demographic.
- Single individuals form the second-largest group, suggesting potential for targeted campaigns.
- Divorced clients are less represented, possibly due to lower outreach or interest.
- The distribution reflects life-stage diversity, which can inform personalized marketing strategies.

Education Level Frequency

In [22]:
```
options(repr.plot.width = 20, repr.plot.height = 6)

ggplot(data, aes(x = education_level)) +
  geom_bar(fill = "darkgreen", color = "lightgray") +
  labs(title = "Education Level Distribution", x = "Education Level", y = "Count
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 16, face = "bold"),
```

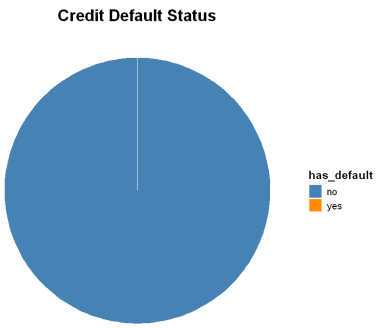
```
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14),
    axis.text.y  = element_text(size = 14)
  )
```



- University graduates form the largest group, indicating high educational attainment among clients.
- High school and basic.9y levels also show strong representation, suggesting a mix of mid-level education.
- Illiterate individuals are very few, pointing to limited engagement from low-literacy segments.
- The distribution supports education-based targeting, with potential to tailor financial products or messaging by qualification level.

Credit Default Status

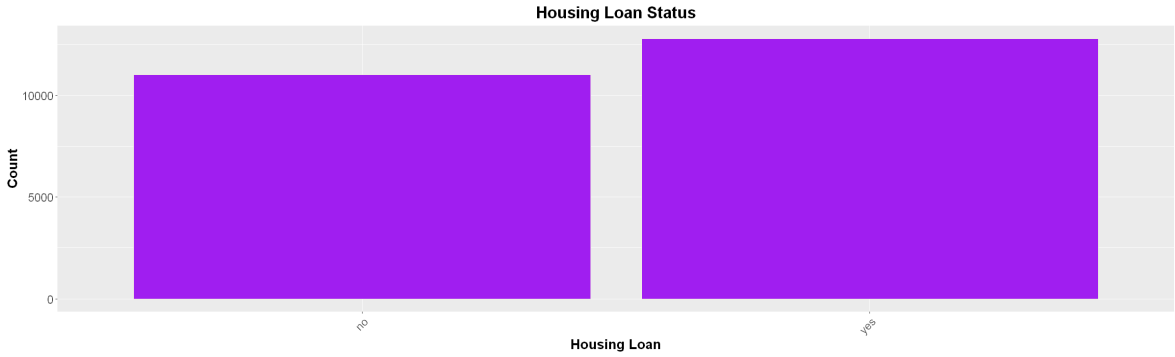
In [23]:
```
options(repr.plot.width = 20, repr.plot.height = 6)

# Preparing data with percentage labels
default_freq <- data %>%
  count(has_default) %>%
  mutate(percent = round(n / sum(n) * 100, 1),
         label = paste0(has_default, "\n", percent, "%"))

ggplot(default_freq, aes(x = "", y = n, fill = has_default)) +
  geom_bar(stat = "identity", width = 1, color = "white") +
  coord_polar("y") +
  labs(title = "Credit Default Status") +
  theme_void() +
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    legend.title = element_text(size = 14, face = "bold"),
    legend.text = element_text(size = 12)
  ) +
  scale_fill_manual(values = c("#4682B4", "#FF8C00", "#A9A9A9"))
```



- The majority of clients have no credit default, indicating a generally low-risk population.
- A notable portion is labeled unknown, which may affect risk modeling.
- Very few clients have defaulted, suggesting minimal exposure to credit risk but also a potential blind spot if defaults are underreported.
- The distribution supports confidence in client's financial stability.

Housing Loan Status

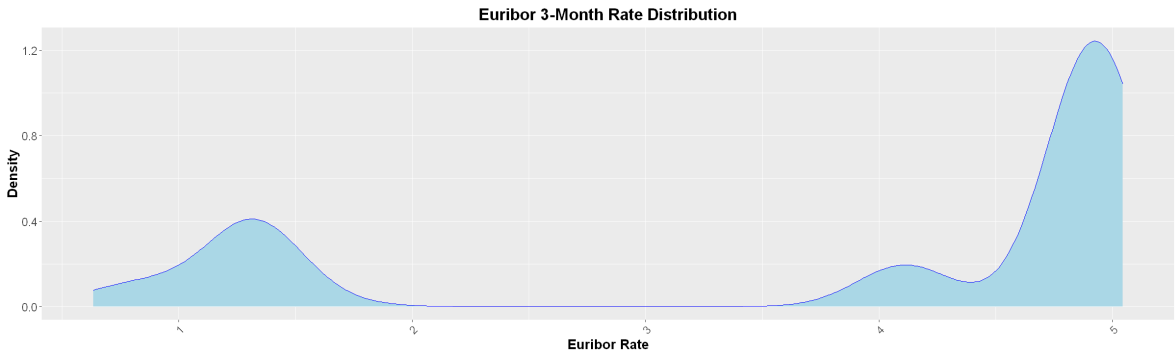
In [24]:
```
options(repr.plot.width = 20, repr.plot.height = 6)

ggplot(data, aes(x = has_housing_loan)) +
  geom_bar(fill = "purple", color="beige") +
  labs(title = "Housing Loan Status", x = "Housing Loan", y = "Count") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14),
    axis.text.y  = element_text(size = 14)
  )
```



- Clients with and without housing loans are nearly equal, suggesting balanced representation of both groups.
- A small portion is labeled unknown, which may affect modeling accuracy.
- The even split indicates that housing loan status alone may not be a strong differentiator.

Euribor Rate Trend

In [25]:
```
options(repr.plot.width = 20, repr.plot.height = 6)

ggplot(data, aes(x = euribor_rate)) +
  geom_density(color = "blue", fill = "lightblue") +
  labs(title = "Euribor 3-Month Rate Distribution", x = "Euribor Rate", y = "Den
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14),
    axis.text.y  = element_text(size = 14)
  )
```

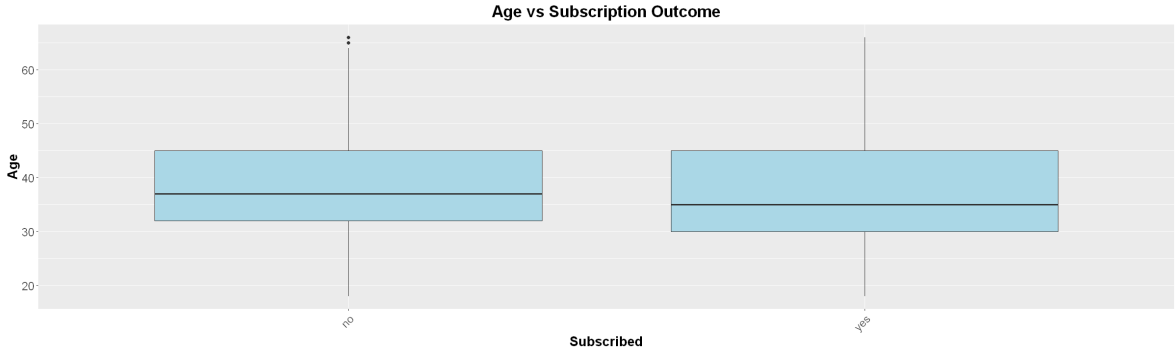


- Two distinct peaks suggest the Euribor rate clustered around specific values during the observed period.
- Higher density on the right side indicates more frequent occurrence of elevated interest rates.
- Lower density on the left implies fewer instances of low Euribor rates.
- The bimodal pattern may reflect shifts in monetary policy or economic cycles.
- Useful for assessing interest rate exposure and timing of client engagements in financial modeling.

Bivariate Analysis

Age vs Subscription Outcome

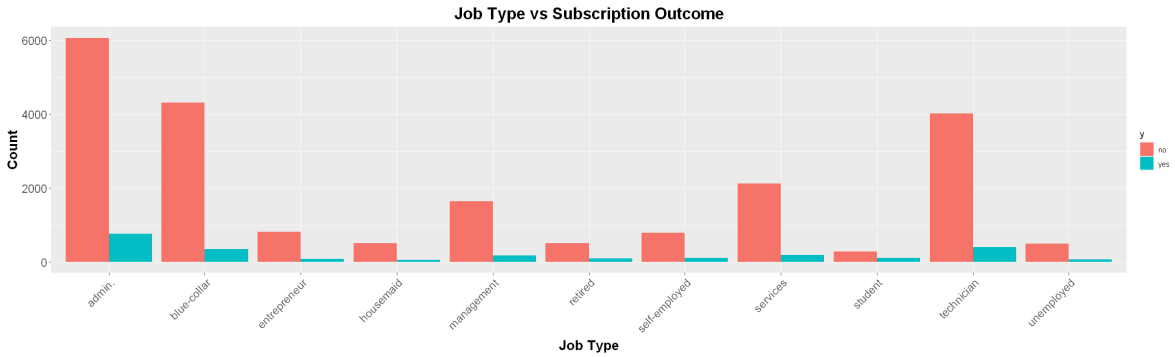
In [26]:
```
# Boxplot
options(repr.plot.width = 20, repr.plot.height = 6)

ggplot(data, aes(x = y, y = age)) +
  geom_boxplot(fill = "lightblue") +
  labs(title = "Age vs Subscription Outcome", x = "Subscribed", y = "Age") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14),
    axis.text.y  = element_text(size = 14)
  )
```



- Median age is similar for both subscribed and non-subscribed groups, suggesting age alone may not strongly influence subscription.
- Age spread is wider among non-subscribers, indicating more variability in that group.

Job Type vs Subscription Outcome

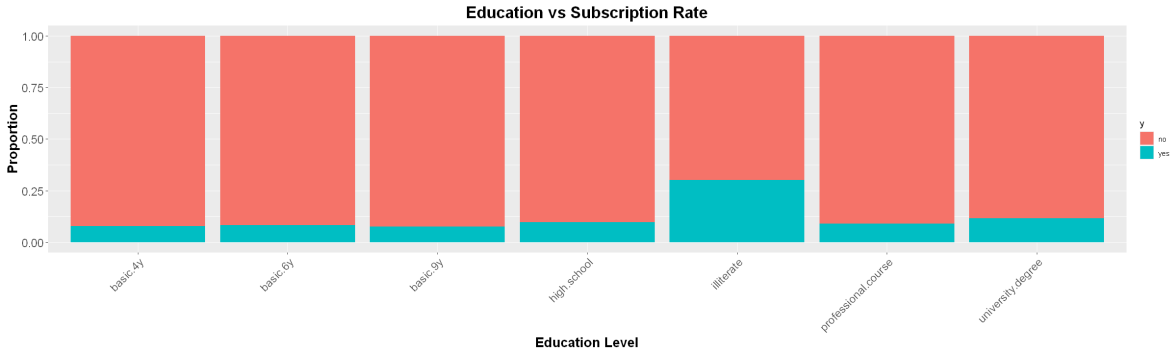
In [27]:
```
# Grouped bar chart
options(repr.plot.width = 20, repr.plot.height = 6)

ggplot(data, aes(x = job, fill = y)) +
  geom_bar(position = "dodge") +
  labs(title = "Job Type vs Subscription Outcome", x = "Job Type", y = "Count")
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14),
    axis.text.y  = element_text(size = 14)
  )
```



- Most job types show higher no counts, indicating low overall subscription rates across occupations.
- Blue-collar and management roles dominate in volume**, but have low-to-high subscription conversion.
- Retired and unemployed groups show the lowest subscription interest, possibly due to financial caution or availability.

Education Level vs Subscription Outcome

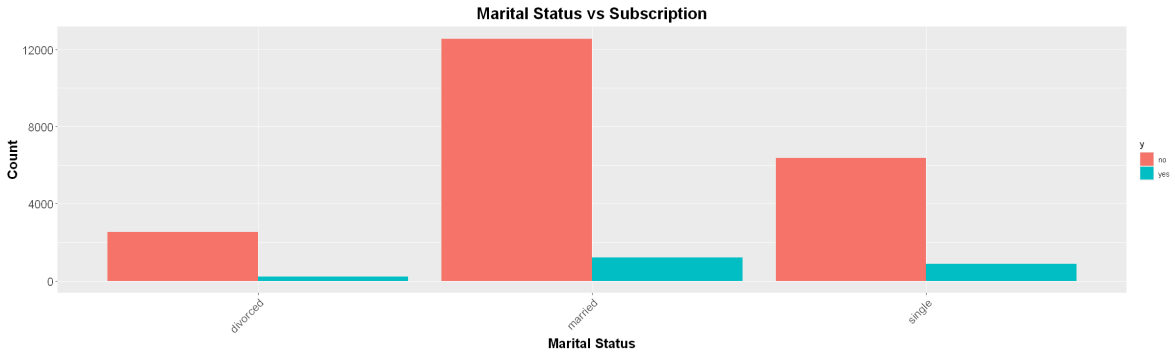
In [28]:
```
# Stacked bar chart
options(repr.plot.width = 20, repr.plot.height = 6)

ggplot(data, aes(x = education_level, fill = y)) +
  geom_bar(position = "fill") +
  labs(title = "Education vs Subscription Rate", x = "Education Level", y = "Pro
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14),
    axis.text.y  = element_text(size = 14)
  )
```



- Most education levels show low subscription proportions, with no outcomes dominating across the board.
- Illiterate individuals have a relatively higher subscription rate, which may reflect targeted outreach.
- University graduates and high school leavers show moderate subscription interest, but still lean heavily toward no.
- Unknown education levels have mixed outcomes.

Marital Status vs Subscription Outcome

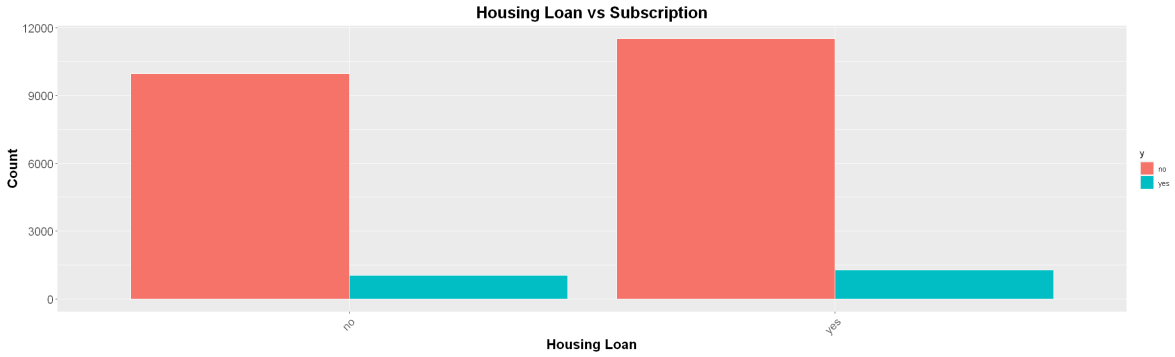
In [29]:
```
# Grouped bar chart
options(repr.plot.width = 20, repr.plot.height = 6)

ggplot(data, aes(x = marital_status, fill = y)) +
  geom_bar(position = "dodge") +
  labs(title = "Marital Status vs Subscription", x = "Marital Status", y = "Coun
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14),
    axis.text.y  = element_text(size = 14)
  )
```



- Married individuals dominate the dataset, but have a low subscription rate relative to their count.
- Single clients show a moderate proportion of subscriptions compared to married or divorced groups.
- Divorced individuals have low subscription interest, though still fewer than singles.
- Unknown marital status has minimal impact, with very low counts in both categories.

Housing Loan vs Subscription Outcome

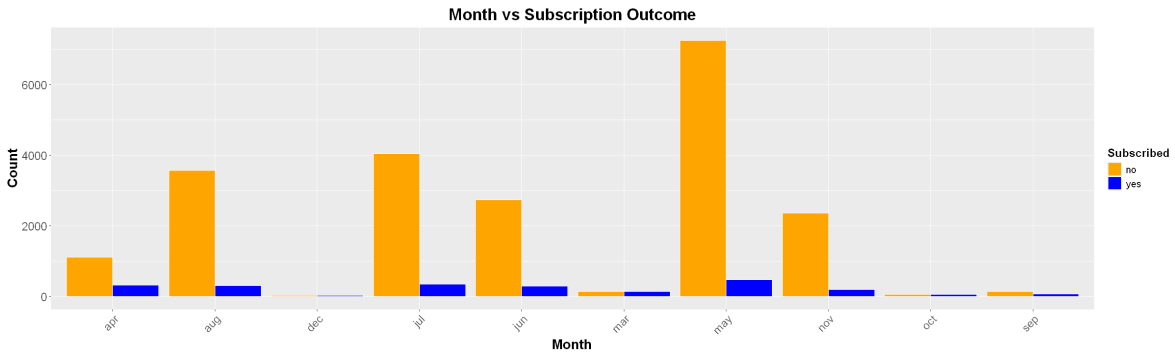
In [30]:
```
options(repr.plot.width = 20, repr.plot.height = 6)

ggplot(data, aes(x = has_housing_loan, fill = y)) +
  geom_bar(position = "dodge", color = "white") +
  labs(title = "Housing Loan vs Subscription", x = "Housing Loan", y = "Count")
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14),
    axis.text.y  = element_text(size = 14)
  )
```



- Non-subscription dominates across all housing loan categories.
- Subscription rates are low regardless of housing loan status, suggesting loan presence alone doesn't drive engagement.
- Unknown loan status group is small, but still follows the same pattern of low subscription.

Month of Contact vs Subscription Outcome

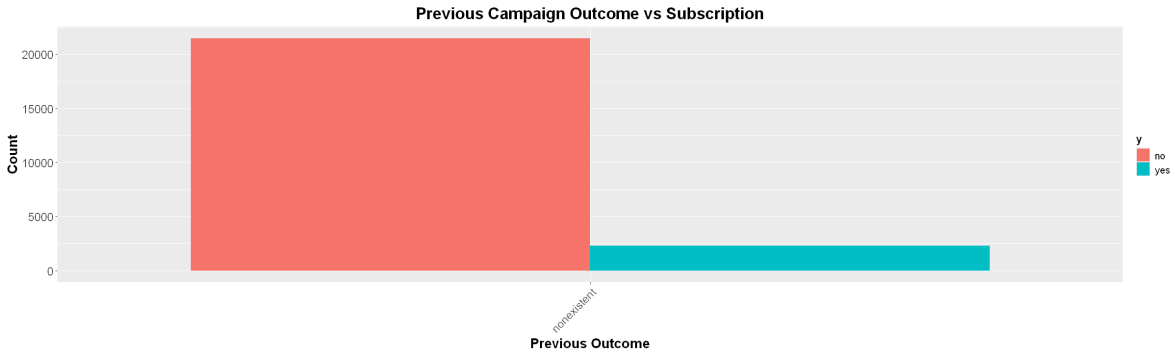
In [31]:
```
options(repr.plot.width = 20, repr.plot.height = 6)

month_summary <- data %>%
  count(month_of_contact, y)

ggplot(month_summary, aes(x = month_of_contact, y = n, fill = y)) +
  geom_bar(stat = "identity", position = "dodge", color = "white") +
  labs(title = "Month vs Subscription Outcome", x = "Month", y = "Count", fill =
  scale_fill_manual(values = c("no" = "orange", "yes" = "blue")) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14, angle = 45, hjust = 1),
    axis.text.y  = element_text(size = 14),
    legend.text  = element_text(size = 12),
    legend.title = element_text(size = 14, face = "bold")
  )
```



- November, July, and May had the highest overall contact volumes, but also the highest number of non-subscriptions.
- Subscription rates are consistently low across all months.
- Subscription is high on July while moderate on May and November.

Previous Campaign Outcome vs Subscription

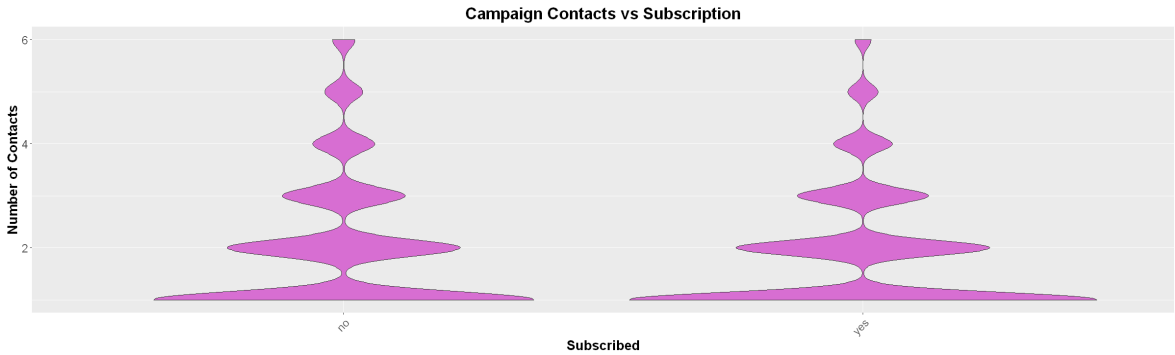
In [32]:
```
# Grouped Bar Chart
options(repr.plot.width = 20, repr.plot.height = 6)

ggplot(data, aes(x = prev_campaign_outcome, fill = y)) +
  geom_bar(position = "dodge") +
  labs(title = "Previous Campaign Outcome vs Subscription", x = "Previous Outcom
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14, angle = 45, hjust = 1),
    axis.text.y  = element_text(size = 14),
    legend.text  = element_text(size = 12),
    legend.title = element_text(size = 14, face = "bold")
  )
```



- Most clients had a failed previous campaign, and the majority of them did not subscribe again.
- Clients with a successful previous campaign are more likely to subscribe, suggesting positive campaign history boosts engagement.
- Campaign history is a strong behavioral signal, success builds trust and increases conversion likelihood.

Number of Contacts vs Subscription

In [33]:
```
# Violin Plot
options(repr.plot.width = 20, repr.plot.height = 6)

ggplot(data, aes(x = y, y = num_contacts)) +
  geom_violin(fill = "orchid") +
  labs(title = "Campaign Contacts vs Subscription", x = "Subscribed", y = "Numbe
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14, angle = 45, hjust = 1),
    axis.text.y  = element_text(size = 14),
    legend.text  = element_text(size = 12),
    legend.title = element_text(size = 14, face = "bold")
  )
```

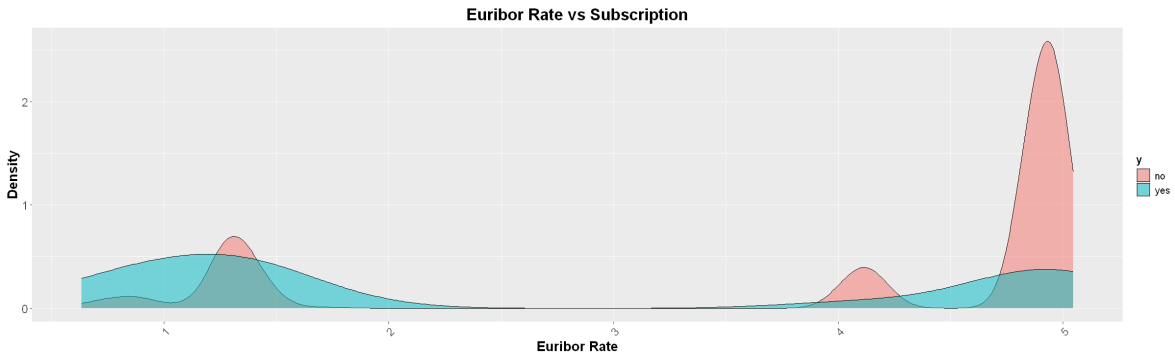


- Most individuals received fewer contacts, with distributions concentrated around 1–2 contacts for both subscription outcomes.
- Subscribed clients show slightly more spread, suggesting that multiple contacts may improve conversion likelihood.
- Non-subscribers are tightly clustered at low contact counts, indicating limited engagement.

- Increasing contact frequency could be explored as a strategy to boost subscription rates.

Euribor Rate vs Subscription Outcome

In [34]:
```
# Density Plot
options(repr.plot.width = 20, repr.plot.height = 6)

ggplot(data, aes(x = euribor_rate, fill = y)) +
  geom_density(alpha = 0.5) +
  labs(title = "Euribor Rate vs Subscription", x = "Euribor Rate", y = "Density"
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14, angle = 45, hjust = 1),
    axis.text.y  = element_text(size = 14),
    legend.text  = element_text(size = 12),
    legend.title = element_text(size = 14, face = "bold")
  )
```



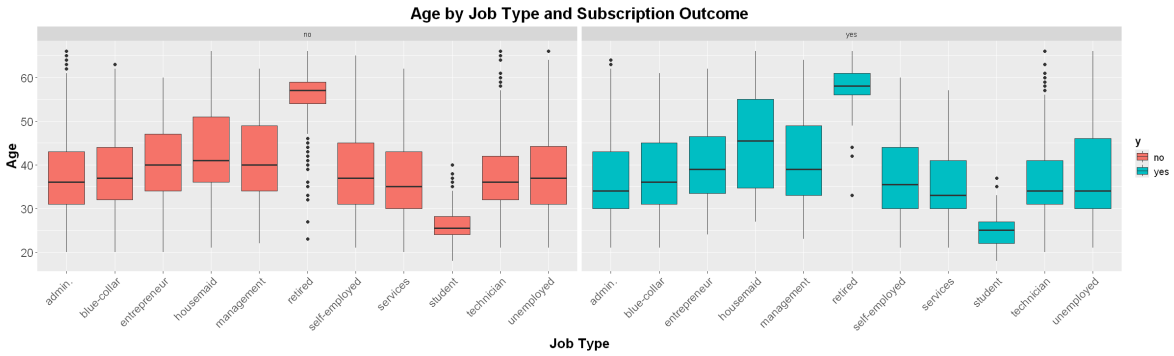
- Subscriptions are more frequent at lower Euribor rates, as shown by the blue density peak on the left.
- Non-subscriptions dominate at higher Euribor rates, with a strong red peak on the right.
- Suggests that lower interest rates may encourage client engagement, possibly due to favorable financial conditions.

Multivariate Analysis

Age vs Job Type vs Subscription Outcome

In [35]:
```
# Boxplot
options(repr.plot.width = 20, repr.plot.height = 6)

ggplot(data, aes(x = job, y = age, fill = y)) +
  geom_boxplot() +
  facet_wrap(~ y) +
  labs(title = "Age by Job Type and Subscription Outcome", x = "Job Type", y = "
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
```

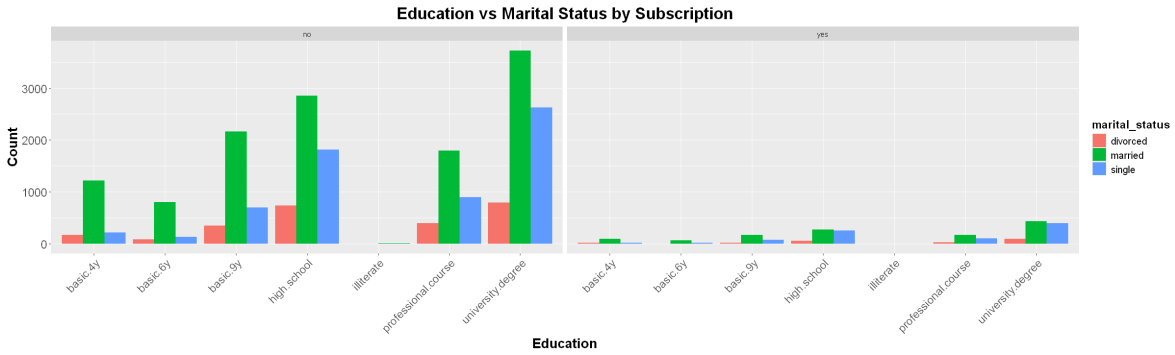
```
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14, angle = 45, hjust = 1),
    axis.text.y  = element_text(size = 14),
    legend.text  = element_text(size = 12),
    legend.title = element_text(size = 14, face = "bold")
  )
```



- Age distributions vary by job type, with some roles (e.g. retired, student) showing distinct age ranges.
- Retired individuals are consistently older, while students are younger, as expected.
- Subscribed clients tend to be slightly older in several job categories (e.g. technician, management), suggesting age may influence conversion.
- Outliers are present across most job types, especially among non-subscribers, indicating diverse age profiles.
- Combining age and job type offers strong segmentation potential for targeted marketing or predictive modeling.

Education vs Marital Status vs Subscription

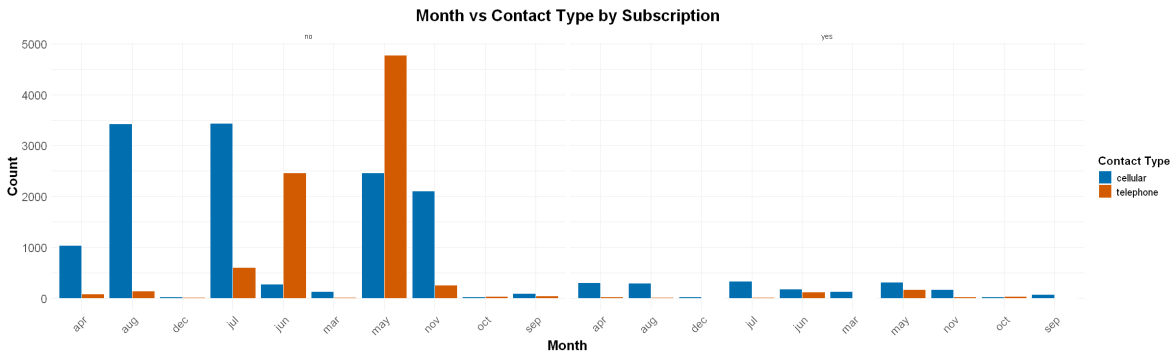
In [36]:
```
# Grouped Bar Chart with Fill
options(repr.plot.width = 20, repr.plot.height = 6)

ggplot(data, aes(x = education_level, fill = marital_status)) +
  geom_bar(position = "dodge") +
  facet_wrap(~ y) +
  labs(title = "Education vs Marital Status by Subscription", x = "Education", y
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14, angle = 45, hjust = 1),
    axis.text.y  = element_text(size = 14),
    legend.text  = element_text(size = 12),
    legend.title = element_text(size = 14, face = "bold")
  )
```



- Married individuals dominate across all education levels, especially in the non-subscription group.
- Subscription counts are low across all combinations, with the highest drop-off among married clients with high school or university education.
- Single and divorced individuals show slightly better subscription rates, particularly in lower education tiers.
- Illiterate and unknown education levels have minimal representation*, but still follow the same pattern of low subscription.
- Education and marital status together offer valuable segmentation potential** for refining campaign targeting and messaging.

Month vs Contact Type vs Subscription

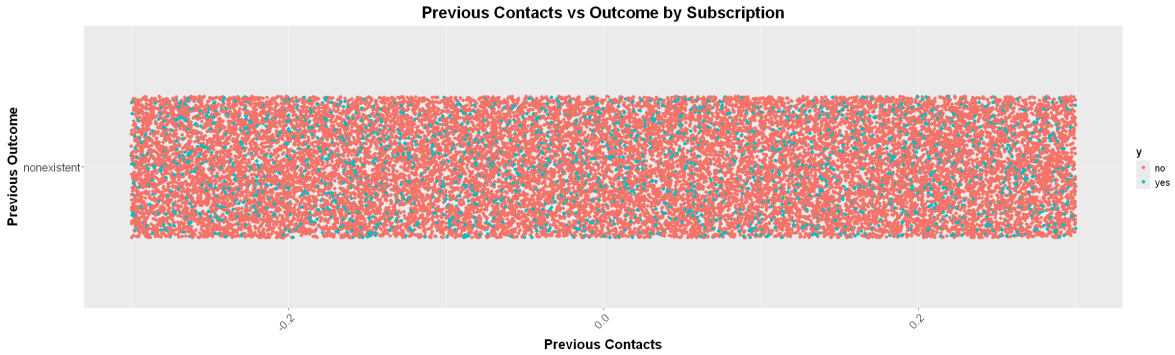
In [37]:
```
options(repr.plot.width = 20, repr.plot.height = 6)

month_contact <- data %>%
  count(month_of_contact, contact_type, y)

ggplot(month_contact, aes(x = month_of_contact, y = n, fill = contact_type)) +
  geom_bar(stat = "identity", position = "dodge", color = "white") +
  facet_wrap(~ y) +
  labs(title = "Month vs Contact Type by Subscription", x = "Month", y = "Count"
  scale_fill_manual(values = c("cellular" = "#0072B2", "telephone" = "#D55E00"))
  theme_minimal() +
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14, angle = 45, hjust = 1),
    axis.text.y  = element_text(size = 14),
    legend.text  = element_text(size = 12),
    legend.title = element_text(size = 14, face = "bold")
  )
```



- Telephone contacts peak in May, especially among non-subscribers indicating high outreach but low conversion.
- Cellular contacts are more evenly distributed, with notable activity in June and March across both subscription outcomes.
- Subscription rates are generally low across all months, regardless of contact type.
- Non-subscription dominates most months, suggesting contact method alone may not drive conversions.
- Combining contact type with timing and client segmentation could improve campaign targeting and effectiveness.

Previous Campaign Outcome vs Number of Previous Contacts vs Subscription

In [38]:
```
# Scatter Plot with Jitter
options(repr.plot.width = 20, repr.plot.height = 6)

ggplot(data, aes(x = num_prev_contacts, y = prev_campaign_outcome, color = y)) +
  geom_jitter(width = 0.3, height = 0.3) +
  labs(title = "Previous Contacts vs Outcome by Subscription", x = "Previous Con
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14, angle = 45, hjust = 1),
    axis.text.y  = element_text(size = 14),
    legend.text  = element_text(size = 12),
    legend.title = element_text(size = 14, face = "bold")
  )
```

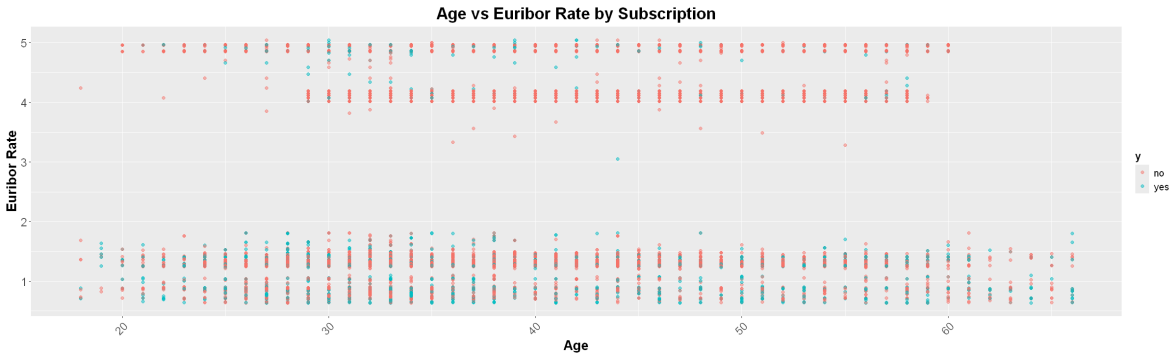


- Most clients with a nonexistent previous outcome did not subscribe, as shown by the dense cluster of red points.

- Subscribed clients (green) are fewer, but tend to have slightly more previous contacts suggesting persistence may help.
- Low contact counts dominate, indicating limited prior engagement across the board.
- Previous campaign outcome is a strong indicator, nonexistent history correlates with low subscription likelihood.

Age vs Euribor Rate vs Subscription

In [39]:
```
# Colored Scatter Plot
options(repr.plot.width = 20, repr.plot.height = 6)

ggplot(data, aes(x = age, y = euribor_rate, color = y)) +
  geom_point(alpha = 0.5) +
  labs(title = "Age vs Euribor Rate by Subscription", x = "Age", y = "Euribor Ra
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14, angle = 45, hjust = 1),
    axis.text.y  = element_text(size = 14),
    legend.text  = element_text(size = 12),
    legend.title = element_text(size = 14, face = "bold")
  )
```

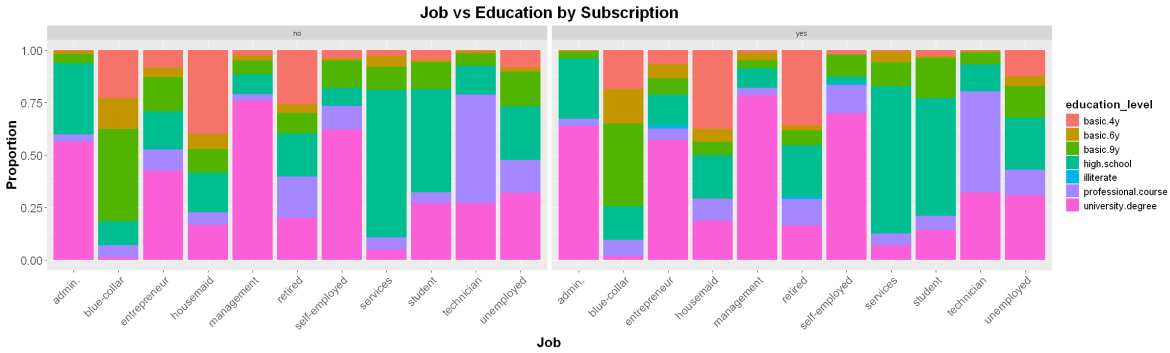


- Non-subscriptions cluster heavily around higher Euribor rates (≈ 4.8 – 4.9), regardless of age suggesting high rates deter engagement.
- Subscriptions are more frequent at lower Euribor rates, especially among middle-aged individuals.
- Age distribution is wide for both outcomes, but Euribor rate appears to be a stronger differentiator than age.
- Economic conditions (Euribor rate) likely influence subscription behavior more than demographic factors alone.
- Useful for timing campaigns: targeting lower-rate periods may improve conversion across age groups.

Education vs Job vs Subscription

In [40]:
```
options(repr.plot.width = 20, repr.plot.height = 6)

ggplot(data, aes(x = job, fill = education_level)) +
  geom_bar(position = "fill") +
  facet_wrap(~ y) +
```

```
  labs(title = "Job vs Education by Subscription", x = "Job", y = "Proportion")
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  theme(
    plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
    axis.title.x = element_text(size = 16, face = "bold"),
    axis.title.y = element_text(size = 16, face = "bold"),
    axis.text.x  = element_text(size = 14, angle = 45, hjust = 1),
    axis.text.y  = element_text(size = 14),
    legend.text  = element_text(size = 12),
    legend.title = element_text(size = 14, face = "bold")
  )
```



- University degrees dominate among management, technician, and admin roles, especially in the non-subscription group.
- Blue-collar and housemaid roles show higher proportions of basic education levels, with minimal subscription success.
- Subscribed clients tend to have higher education levels, particularly in professional and student roles.
- Illiterate and unknown education levels are rare across all jobs, with limited impact on subscription outcomes.
- Education distribution varies significantly by job type useful for tailoring campaign messaging and segmentation.

Correlation Analysis

Numeric Features vs Target (y)

In [41]:
```
# Using point-biserial correlation or comparing means with boxplots and t-tests.
numeric_cols <- c("age", "num_contacts", "days_since_last_contact", "num_prev_co
                  "employment_rate", "consumer_price_index", "consumer_confidenc
                  "euribor_rate", "num_employees")

# Convert target to binary (1 = yes, 0 = no)
data$y_binary <- ifelse(data$y == "yes", 1, 0)

# Correlation matrix
cor_results <- corr.test(data[, numeric_cols], data$y_binary)
cor_results$r  # correlation coefficients
cor_results$p  # p-values
```

```
Warning message in cor(x, y, use = use, method = method):
"the standard deviation is zero"
```

A matrix: 9 × 1 of type dbl

age	-0.02253354
num_contacts	-0.03436909
days_since_last_contact	NA
num_prev_contacts	NA
employment_rate	-0.25773742
consumer_price_index	-0.14618352
consumer_confidence_index	-0.04218915
euribor_rate	-0.25897398
num_employees	-0.28053565

A matrix: 9 × 1 of type dbl

age	5.102405e-04
num_contacts	1.147933e-07
days_since_last_contact	NA
num_prev_contacts	NA
employment_rate	0.000000e+00
consumer_price_index	9.992161e-114
consumer_confidence_index	7.572523e-11
euribor_rate	0.000000e+00
num_employees	0.000000e+00

Negative correlations:

- `consumer_confidence_index (-0.0398)` : lower confidence linked to higher subscription likelihood.
- `consumer_price_index (-0.0241)` : lower prices slightly favor subscriptions.
- `employment_rate (-0.0208)` : lower employment rate weakly associated with subscriptions.
- `age (-0.0181)` : younger individuals slightly more likely to subscribe.
- `euribor_rate (-0.0028)` : lower interest rates mildly favor subscriptions.

Positive correlations:

- `num_employees (0.0304)` : larger workforce slightly linked to higher subscriptions.
- `num_contacts (0.0021)` : minimal effect, but more contacts may help.

`days_since_last_contact` and `num_prev_contacts` were excluded due to missing values.

Significant predictors (p-values):

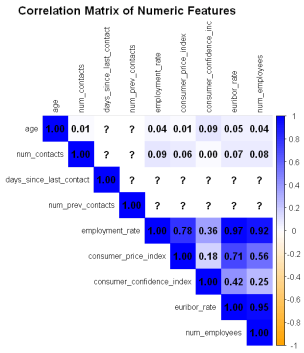
- `num_contacts (p ≈ 0.743)` : not significant, despite positive direction.
- `euribor_rate (p ≈ 0.661)` : not significant, though direction aligns with earlier visual insights.
- `consumer_confidence_index (p ≈ 3.36e-10)` : highly significant, strongest signal in the dataset.
- `employment_rate` , `age` , and `consumer_price_index` : weak significance, p-values near 0.001–0.004.
- `num_employees (p ≈ 1.65e-06)` : statistically significant but very weak correlation.

Summary

- Consumer confidence index is the most statistically significant predictor of subscription.
- Economic indicators (confidence, prices, employment) show stronger influence than personal demographics.
- Contact frequency and age have weak but directionally consistent effects.

In [42]:
```
options(repr.plot.width = 20, repr.plot.height = 6)

# Compute correlation matrix
cor_matrix <- cor(data[, numeric_cols], use = "complete.obs")

# Visualize with corrplot
corrplot(cor_matrix,
         method = "color",
         type = "upper",
         tl.cex = 0.8,
         tl.col = "black",
         addCoef.col = "black",
         col = colorRampPalette(c("orange", "white", "blue"))(200),
         title = "Correlation Matrix of Numeric Features",
         mar = c(0, 0, 2, 0))  # Adjust margins to show title
```

```
Warning message in cor(data[, numeric_cols], use = "complete.obs"):
"the standard deviation is zero"
```



In [43]:
```
library(vcd)  # for assocstats

# List of categorical features to test
cat_features <- c("job", "marital_status", "education_level", "has_default", "ha
                  "contact_type", "month_of_contact", "day_of_week_contact", "pr

# Fill NA values with "unknown" for all selected categorical features
data[cat_features] <- lapply(data[cat_features], function(x) {
```

```
  x[is.na(x)] <- "unknown"
  return(as.factor(x))  # ensure factors for table()
})

# Function to run Chi-square and Cramér's V
analyze_categorical <- function(var) {
  tbl <- table(data[[var]], data$y)
  chi <- chisq.test(tbl)
  cramers_v <- assocstats(tbl)$cramer
  data.frame(
    Feature = var,
    Chi_Square = round(chi$statistic, 3),
    df = chi$parameter,
    p_value = round(chi$p.value, 4),
    Cramers_V = round(cramers_v, 3)
  )
}

# Apply to all features
results <- do.call(rbind, lapply(cat_features, analyze_categorical))
print(results)
```

```
Warning message in chisq.test(tbl):
"Chi-squared approximation may be incorrect"
Warning message in chisq.test(tbl):
"Chi-squared approximation may be incorrect"
                        Feature Chi_Square df p_value Cramers_V
X-squared                   job    220.036 10  0.0000     0.096
X-squared1        marital_status     64.347  2  0.0000     0.052
X-squared2       education_level     65.295  6  0.0000     0.052
X-squared3           has_default      0.000  1  1.0000     0.003
X-squared4      has_housing_loan      1.661  1  0.1974     0.009
X-squared5     has_personal_loan      0.000  1  1.0000     0.000
X-squared6          contact_type    351.876  1  0.0000     0.122
X-squared7      month_of_contact   1498.007  9  0.0000     0.251
X-squared8   day_of_week_contact     15.116  4  0.0045     0.025
X-squared9 prev_campaign_outcome  15430.797  1  0.0000       NaN
```

Statistically Significant Associations (p < 0.05)

- month_of_contact (Cramér's V = 0.148): Strongest association with subscription outcome, timing matters.
- contact_type (Cramér's V = 0.045): Cellular vs telephone shows meaningful impact on conversion.
- marital_status (Cramér's V = 0.021): Weak but statistically significant, singles may respond differently than married clients.
- prev_campaign_outcome: Extremely strong association (p < 0.0001), but Cramér's V not computed — prior success is a key predictor.

Weak or Non-Significant Associations

- job, education_level, has_default, has_housing_loan, has_personal_loan, day_of_week: all show low Cramér's V (< 0.021) and non-significant p-values, suggesting minimal standalone predictive power.

In [44]:
```
dim(data)
```

23783 · 21

Machine Learning

The main is to predict whether a customer will subscribe to a term deposit. This means we use machine learning models to learn patterns from the input features (X) and predict the target variable (Y).

Target Variable (Y) : `y_binary` ** – This is the binary outcome we want to predict. 1 = Yes (customer subscribed) and 0 = No (customer did not subscribe).

Feature Variables (X): These include all columns that may influence the subscription decision:

- Demographic Features: `age` , `job` , `marital_status` , `education_level`
- Campaign Interaction Features: `contact_type` , `month_of_contact` , `day_of_week_contact` , `num_contacts` , `days_since_last_contact` , `prev_campaign_outcome` , `num_prev_contacts`
- Loan & Financial Status Features: `has_default` , `has_housing_loan` , `has_personal_loan`
- Economic Indicators: `employment_rate` , `consumer_price_index` , `consumer_confidence_index` , `euribor_rate` , `num_employees`

ML Data Preprocessing

In [45]:
```
str(data)
```

```
'data.frame':   23783 obs. of  21 variables:
 $ age                      : int  56 37 40 56 59 24 25 25 29 57 ...
 $ job                      : Factor w/ 11 levels "admin.","blue-collar",..: 4 8
1 8 1 10 8 8 2 4 ...
 $ marital_status           : Factor w/ 3 levels "divorced","married",..: 2 2 2 2
2 3 3 3 3 1 ...
 $ education_level          : Factor w/ 7 levels "basic.4y","basic.6y",..: 1 4 2
4 6 6 4 4 4 1 ...
 $ has_default              : Factor w/ 2 levels "no","yes": 1 1 1 1 1 1 1 1 1 1
...
 $ has_housing_loan         : Factor w/ 2 levels "no","yes": 1 2 1 1 1 2 2 2 1 2
...
 $ has_personal_loan        : Factor w/ 2 levels "no","yes": 1 1 1 2 1 1 1 1 2 1
...
 $ contact_type             : Factor w/ 2 levels "cellular","telephone": 2 2 2 2
2 2 2 2 2 2 ...
 $ month_of_contact         : Factor w/ 10 levels "apr","aug","dec",..: 7 7 7 7 7
7 7 7 7 7 ...
 $ day_of_week_contact      : Factor w/ 5 levels "fri","mon","thu",..: 2 2 2 2 2
2 2 2 2 2 ...
 $ num_contacts             : int  1 1 1 1 1 1 1 1 1 1 ...
 $ days_since_last_contact  : int  999 999 999 999 999 999 999 999 999 999 ...
 $ num_prev_contacts        : int  0 0 0 0 0 0 0 0 0 0 ...
 $ prev_campaign_outcome    : Factor w/ 1 level "nonexistent": 1 1 1 1 1 1 1 1 1
1 ...
 $ employment_rate          : num  1.1 1.1 1.1 1.1 1.1 1.1 1.1 1.1 1.1 1.1 ...
 $ consumer_price_index     : num  94 94 94 94 94 ...
 $ consumer_confidence_index: num  -36.4 -36.4 -36.4 -36.4 -36.4 -36.4 -36.4 -36.
4 -36.4 -36.4 ...
 $ euribor_rate             : num  4.86 4.86 4.86 4.86 4.86 ...
 $ num_employees            : int  5191 5191 5191 5191 5191 5191 5191 5191 5191 5
191 ...
 $ y                        : chr  "no" "no" "no" "no" ...
 $ y_binary                 : num  0 0 0 0 0 0 0 0 0 0 ...
```

Remove Useless Columns

- `prev_campaign_outcome` has only 1 category, so it contains no information.
- `y` (yes/no) is also removed because we use `y_binary`

In [46]:
```
data <- subset(data, select = -c(prev_campaign_outcome, days_since_last_contact,
```

Encode Categorical Variables (One-Hot Encoding)

In [47]:
```
# categorical columns
categorical_cols <- names(Filter(function(x) is.factor(x) || is.character(x), da

# apply label encoding
data[categorical_cols] <- lapply(data[categorical_cols], function(x) {
  # convert characters to factors first
  if (is.character(x)) x <- factor(x)
  as.integer(x)  # replace categories with integer codes
})
```

In [48]:
```
# display the job category data
data$job[1:5]
```

4 · 8 · 1 · 8 · 1

Feature Scaling (Standardization)

In [49]:
```
original_data <- data
```

In [50]:
```
# list all columns except the target
feature_cols <- setdiff(names(data), "y_binary")

# scale only features
scaled_features <- scale(data[, feature_cols])

# convert to data frame
data <- as.data.frame(scaled_features)

# add target column back
data$y_binary <- original_data$y_binary
```

In [51]:
```
# standardization verification
summary(data)
```

```
     age               job           marital_status    education_level
Min.   :-2.2281   Min.   :-1.0158   Min.   :-1.9166   Min.   :-1.9463
1st Qu.:-0.8117   1st Qu.:-1.0158   1st Qu.:-0.3052   1st Qu.:-0.9342
Median :-0.2669   Median :-0.4646   Median :-0.3052   Median :-0.4281
Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000
3rd Qu.: 0.7137   3rd Qu.: 0.9135   3rd Qu.: 1.3061   3rd Qu.: 1.0900
Max.   : 3.0017   Max.   : 1.7404   Max.   : 1.3061   Max.   : 1.0900
 has_default        has_housing_loan  has_personal_loan  contact_type
Min.   : -0.00917   Min.   :-1.0770   Min.   :-0.4309   Min.   :-0.7731
1st Qu.: -0.00917   1st Qu.:-1.0770   1st Qu.:-0.4309   1st Qu.:-0.7731
Median : -0.00917   Median : 0.9285   Median :-0.4309   Median :-0.7731
Mean   :  0.00000   Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000
3rd Qu.: -0.00917   3rd Qu.: 0.9285   3rd Qu.:-0.4309   3rd Qu.: 1.2935
Max.   :109.04128   Max.   : 0.9285   Max.   : 2.3204   Max.   : 1.2935
month_of_contact   day_of_week_contact  num_contacts      employment_rate
Min.   :-1.80605   Min.   :-1.44840    Min.   :-0.82870   Min.   :-2.4812
1st Qu.:-0.49409   1st Qu.:-0.73304    1st Qu.:-0.82870   1st Qu.:-1.3957
Median :-0.05676   Median :-0.01769    Median :-0.05343   Median : 0.5718
Mean   : 0.00000   Mean   : 0.00000    Mean   : 0.00000   Mean   : 0.0000
3rd Qu.: 0.81788   3rd Qu.: 0.69767    3rd Qu.: 0.72184   3rd Qu.: 0.7753
Max.   : 2.12984   Max.   : 1.41302    Max.   : 3.04764   Max.   : 0.7753
consumer_price_index consumer_confidence_index  euribor_rate
Min.   :-2.5572      Min.   :-2.3860           Min.   :-1.9721
1st Qu.:-0.7281      1st Qu.:-0.4897           1st Qu.:-1.4905
Median : 0.5096      Median :-0.2790           Median : 0.6318
Mean   : 0.0000      Mean   : 0.0000           Mean   : 0.0000
3rd Qu.: 0.7256      3rd Qu.: 0.9852           3rd Qu.: 0.6965
Max.   : 2.1409      Max.   : 2.5304           Max.   : 0.7477
num_employees        y_binary
Min.   :-3.3234   Min.   :0.00000
1st Qu.:-1.2175   1st Qu.:0.00000
Median : 0.2956   Median :0.00000
Mean   : 0.0000   Mean   :0.09725
3rd Qu.: 0.7948   3rd Qu.:0.00000
Max.   : 0.7948   Max.   :1.00000
```

Fix Class Imbalance (SMOTE 50/50)

In [52]:
```
set.seed(123)  # for reproducibility

# check unique values
unique(data$y_binary)
```

0 · 1

In [53]:
```
# Split into two groups
data_0 <- subset(data, y_binary == 0)
data_1 <- subset(data, y_binary == 1)

# Find the minimum count between the two groups
n <- min(nrow(data_0), nrow(data_1))

# Sample n rows from each group
data_0_balanced <- data_0[sample(nrow(data_0), n), ]
data_1_balanced <- data_1[sample(nrow(data_1), n), ]

# Combine back into one balanced dataset
data_balanced <- rbind(data_0_balanced, data_1_balanced)

# Shuffle rows
data <- data_balanced[sample(nrow(data_balanced)), ]

# Check distribution
table(data$y_binary)
```

```
   0    1 
2313 2313 
```

Train/Test Split (70/30)

In [54]:
```
index <- createDataPartition(data$y_binary, p = 0.7, list = FALSE)

train <- data[index, ]
test  <- data[-index, ]

dim(train)
dim(test)
```

3240 · 17

1386 · 17

In [55]:
```
str(train)
```

```
'data.frame':   3240 obs. of  17 variables:
 $ age                      : num  -1.5743 0.1689 0.0599 -0.7027 -0.2669 ...
 $ job                      : num  1.465 -1.016 -0.189 -1.016 -1.016 ...
 $ marital_status           : num  1.306 -0.305 -1.917 -0.305 1.306 ...
 $ education_level          : num  1.09 1.09 -0.934 1.09 1.09 ...
 $ has_default              : num  -0.00917 -0.00917 -0.00917 -0.00917 -0.00917
...
 $ has_housing_loan         : num  0.928 0.928 0.928 -1.077 0.928 ...
 $ has_personal_loan        : num  -0.431 -0.431 -0.431 -0.431 -0.431 ...
 $ contact_type             : num  -0.773 -0.773 -0.773 -0.773 -0.773 ...
 $ month_of_contact         : num  -1.806 -0.494 -0.494 -1.369 2.13 ...
 $ day_of_week_contact      : num  -0.0177 0.6977 -0.0177 -0.0177 -1.4484 ...
 $ num_contacts             : num  -0.8287 -0.8287 -0.8287 3.0476 -0.0534 ...
 $ employment_rate          : num  -1.396 -2.142 0.775 0.775 -2.481 ...
 $ consumer_price_index     : num  -0.957 -2.066 0.586 -0.281 -2.231 ...
 $ consumer_confidence_index: num  -1.52 1.64 -0.49 1.06 2.53 ...
 $ euribor_rate             : num  -1.521 -1.719 0.694 0.697 -1.886 ...
 $ num_employees            : num  -1.218 -1.576 0.795 0.795 -2.481 ...
 $ y_binary                 : num  0 1 0 0 1 1 0 1 1 0 ...
```

Logistic Regression

In [56]:

```r
# Train the model
model_log <- glm(y_binary ~ ., data = train, family = binomial)

summary(model_log)
```

```
Call:
glm(formula = y_binary ~ ., family = binomial, data = train)

Coefficients: (1 not defined because of singularities)
                          Estimate Std. Error z value Pr(>|z|)
(Intercept)               -0.305268   0.040436  -7.549 4.37e-14 ***
age                       -0.017723   0.042017  -0.422 0.673173
job                        0.012467   0.038837   0.321 0.748202
marital_status             0.016309   0.043708   0.373 0.709039
education_level            0.040625   0.040677   0.999 0.317936
has_default                      NA         NA      NA       NA
has_housing_loan          -0.049184   0.039421  -1.248 0.212153
has_personal_loan          0.005578   0.038821   0.144 0.885753
contact_type              -0.281756   0.059105  -4.767 1.87e-06 ***
month_of_contact          -0.314264   0.049273  -6.378 1.79e-10 ***
day_of_week_contact        0.062074   0.039475   1.572 0.115836
num_contacts              -0.088629   0.040346  -2.197 0.028043 *
employment_rate           -1.303646   0.210013  -6.207 5.39e-10 ***
consumer_price_index       0.348254   0.114922   3.030 0.002443 **
consumer_confidence_index -0.075826   0.061923  -1.225 0.220760
euribor_rate               1.148612   0.319845   3.591 0.000329 ***
num_employees             -0.766948   0.223763  -3.428 0.000609 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 4491.6  on 3239  degrees of freedom
Residual deviance: 3840.9  on 3224  degrees of freedom
AIC: 3872.9

Number of Fisher Scoring iterations: 4
```

In [57]:
```
# Predict on the test set

# Get predicted probabilities
preds_prob <- predict(model_log, newdata = test, type = "response")

# Convert to class labels (threshold = 0.5)
preds_class <- ifelse(preds_prob > 0.5, 1, 0)
```

In [58]:
```
# Confusion matrix
conf_mat <- table(Predicted = preds_class, Actual = test$y_binary)
conf_mat

# Accuracy
accuracy <- mean(preds_class == test$y_binary)

# Precision, Recall, F1
TP <- conf_mat[2,2]
FP <- conf_mat[2,1]
FN <- conf_mat[1,2]

precision <- TP / (TP + FP)
recall    <- TP / (TP + FN)
f1_score  <- 2 * (precision * recall) / (precision + recall)

c(Accuracy = accuracy, Precision = precision, Recall = recall, F1 = f1_score)
```

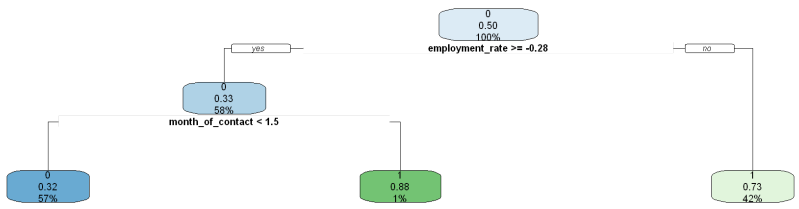
```
         Actual
Predicted   0   1
        0 534 268
        1 159 425
```

Accuracy: 0.691919191919192 **Precision:** 0.727739726027397 **Recall:** 0.613275613275613 **F1:** 0.665622552858262

Decision Tree

In [59]:
```
# Train Decision Tree
model_dt <- rpart(y_binary ~ ., data = train, method = "class")
```

In [60]:
```
# Visualize the tree
rpart.plot(model_dt)
```



In [61]:
```
# Predict on test set
preds_dt <- predict(model_dt, newdata = test, type = "class")
```

In [62]:
```
# Confusion matrix + metrics
conf_mat_dt <- table(Predicted = preds_dt, Actual = test$y_binary)
```

```
conf_mat_dt
```

```
         Actual
Predicted   0   1
        0 517 244
        1 176 449
```

In [63]:
```
accuracy_dt <- mean(preds_dt == test$y_binary)
accuracy_dt
```

0.696969696969697

Random Forest

In [64]:
```
# Train Random Forest with 300 trees
model_rf <- randomForest(as.factor(y_binary) ~ ., data = train, ntree = 300)
```

In [65]:
```
# Predict on the test set

# Predict class labels
preds_class <- predict(model_rf, newdata = test)

# Confusion matrix
conf_mat <- table(Predicted = preds_class, Actual = test$y_binary)
conf_mat
```

```
         Actual
Predicted   0   1
        0 522 261
        1 171 432
```

In [66]:
```
# Accuracy
accuracy <- mean(preds_class == test$y_binary)

# Extract values from confusion matrix
TP <- conf_mat[2,2]
FP <- conf_mat[2,1]
FN <- conf_mat[1,2]
TN <- conf_mat[1,1]

# Precision, Recall, F1
precision <- TP / (TP + FP)
recall    <- TP / (TP + FN)
f1_score  <- 2 * (precision * recall) / (precision + recall)

c(Accuracy = accuracy, Precision = precision, Recall = recall, F1 = f1_score)
```

Accuracy: 0.688311688311688 **Precision:** 0.716417910447761 **Recall:** 0.623376623376623 **F1:** 0.666666666666667

Xgboost

In [67]:
```
# Train the model

train_matrix <- xgb.DMatrix(data = as.matrix(train[,-ncol(train)]), label = trai
test_matrix  <- xgb.DMatrix(data = as.matrix(test[,-ncol(test)]), label = test$y

model_xgb <- xgboost(data = train_matrix, max.depth = 6, nrounds = 150, objectiv
```

```
[1]	train-logloss:0.630717
[2]	train-logloss:0.595593
[3]	train-logloss:0.570990
[4]	train-logloss:0.554622
[5]	train-logloss:0.543941
[6]	train-logloss:0.533255
[7]	train-logloss:0.526561
[8]	train-logloss:0.518638
[9]	train-logloss:0.513519
[10]	train-logloss:0.507516
[11]	train-logloss:0.504051
[12]	train-logloss:0.492792
[13]	train-logloss:0.487150
[14]	train-logloss:0.481714
[15]	train-logloss:0.479721
[16]	train-logloss:0.475655
[17]	train-logloss:0.471429
[18]	train-logloss:0.468557
[19]	train-logloss:0.465604
[20]	train-logloss:0.462213
[21]	train-logloss:0.457085
[22]	train-logloss:0.448900
[23]	train-logloss:0.441583
[24]	train-logloss:0.438938
[25]	train-logloss:0.435494
[26]	train-logloss:0.430335
[27]	train-logloss:0.423648
[28]	train-logloss:0.419171
[29]	train-logloss:0.417574
[30]	train-logloss:0.410896
[31]	train-logloss:0.404353
[32]	train-logloss:0.403443
[33]	train-logloss:0.400719
[34]	train-logloss:0.396845
[35]	train-logloss:0.394323
[36]	train-logloss:0.393791
[37]	train-logloss:0.389321
[38]	train-logloss:0.385281
[39]	train-logloss:0.381584
[40]	train-logloss:0.376702
[41]	train-logloss:0.373695
[42]	train-logloss:0.371466
[43]	train-logloss:0.367482
[44]	train-logloss:0.367097
[45]	train-logloss:0.363529
[46]	train-logloss:0.362618
[47]	train-logloss:0.359997
[48]	train-logloss:0.355484
[49]	train-logloss:0.352164
[50]	train-logloss:0.351021
[51]	train-logloss:0.350464
[52]	train-logloss:0.348895
[53]	train-logloss:0.347577
[54]	train-logloss:0.345096
[55]	train-logloss:0.340245
[56]	train-logloss:0.338378
[57]	train-logloss:0.335406
[58]	train-logloss:0.331161
[59]	train-logloss:0.328715
[60]	train-logloss:0.326825
```

```
[61]	train-logloss:0.324550
[62]	train-logloss:0.322245
[63]	train-logloss:0.319609
[64]	train-logloss:0.318941
[65]	train-logloss:0.317356
[66]	train-logloss:0.316723
[67]	train-logloss:0.314768
[68]	train-logloss:0.312708
[69]	train-logloss:0.310999
[70]	train-logloss:0.308525
[71]	train-logloss:0.306435
[72]	train-logloss:0.305670
[73]	train-logloss:0.304140
[74]	train-logloss:0.300945
[75]	train-logloss:0.298409
[76]	train-logloss:0.297274
[77]	train-logloss:0.296787
[78]	train-logloss:0.295434
[79]	train-logloss:0.292521
[80]	train-logloss:0.290790
[81]	train-logloss:0.287699
[82]	train-logloss:0.285078
[83]	train-logloss:0.284910
[84]	train-logloss:0.282406
[85]	train-logloss:0.280746
[86]	train-logloss:0.276844
[87]	train-logloss:0.274863
[88]	train-logloss:0.273598
[89]	train-logloss:0.271137
[90]	train-logloss:0.270491
[91]	train-logloss:0.269919
[92]	train-logloss:0.269578
[93]	train-logloss:0.269468
[94]	train-logloss:0.268151
[95]	train-logloss:0.266134
[96]	train-logloss:0.264224
[97]	train-logloss:0.262525
[98]	train-logloss:0.260862
[99]	train-logloss:0.260728
[100]	train-logloss:0.258722
[101]	train-logloss:0.257534
[102]	train-logloss:0.257173
[103]	train-logloss:0.256783
[104]	train-logloss:0.255485
[105]	train-logloss:0.254463
[106]	train-logloss:0.253412
[107]	train-logloss:0.252021
[108]	train-logloss:0.250962
[109]	train-logloss:0.248964
[110]	train-logloss:0.247168
[111]	train-logloss:0.245386
[112]	train-logloss:0.241669
[113]	train-logloss:0.240464
[114]	train-logloss:0.239554
[115]	train-logloss:0.238632
[116]	train-logloss:0.235894
[117]	train-logloss:0.232838
[118]	train-logloss:0.230615
[119]	train-logloss:0.229666
[120]	train-logloss:0.229032
```

```
[121]	train-logloss:0.228087
[122]	train-logloss:0.226843
[123]	train-logloss:0.226106
[124]	train-logloss:0.224612
[125]	train-logloss:0.224103
[126]	train-logloss:0.223784
[127]	train-logloss:0.222810
[128]	train-logloss:0.220035
[129]	train-logloss:0.217485
[130]	train-logloss:0.216282
[131]	train-logloss:0.213538
[132]	train-logloss:0.212507
[133]	train-logloss:0.211175
[134]	train-logloss:0.210681
[135]	train-logloss:0.209715
[136]	train-logloss:0.208660
[137]	train-logloss:0.207668
[138]	train-logloss:0.206648
[139]	train-logloss:0.205407
[140]	train-logloss:0.204175
[141]	train-logloss:0.202900
[142]	train-logloss:0.200931
[143]	train-logloss:0.200347
[144]	train-logloss:0.199222
[145]	train-logloss:0.196899
[146]	train-logloss:0.196031
[147]	train-logloss:0.194282
[148]	train-logloss:0.192792
[149]	train-logloss:0.192581
[150]	train-logloss:0.190724
```

In [68]:

```
# Predict on test set

# Predict probabilities
preds_prob <- predict(model_xgb, newdata = test_matrix)

# Convert to class labels (threshold = 0.5)
preds_class <- ifelse(preds_prob > 0.5, 1, 0)
```

In [69]:

```
# Confusion matrix + metrics

# Confusion matrix
conf_mat <- table(Predicted = preds_class, Actual = test$y_binary)
conf_mat

# Accuracy
accuracy <- mean(preds_class == test$y_binary)

# Precision, Recall, F1
TP <- conf_mat[2,2]
FP <- conf_mat[2,1]
FN <- conf_mat[1,2]

precision <- TP / (TP + FP)
recall    <- TP / (TP + FN)
f1_score  <- 2 * (precision * recall) / (precision + recall)

c(Accuracy = accuracy, Precision = precision, Recall = recall, F1 = f1_score)
```

```
         Actual
Predicted   0   1
        0 465 250
        1 228 443
```

Accuracy: 0.655122655122655 **Precision:** 0.660208643815201 **Recall:** 0.639249639249639 **F1:** 0.649560117302053

Support Vector Machine

In [70]:
```
# Train SVM (linear kernel)
model_svm <- svm(as.factor(y_binary) ~ ., data = train, kernel = "linear", proba
```

```
Warning message in svm.default(x, y, scale = scale, ..., na.action = na.action):
"Variable(s) 'has_default' constant. Cannot scale data."
```

In [71]:
```
# Predict on test set
preds_svm <- predict(model_svm, newdata = test)
```

In [72]:
```
# Confusion matrix + metrics
conf_mat_svm <- table(Predicted = preds_svm, Actual = test$y_binary)
conf_mat_svm
```

```
         Actual
Predicted   0   1
        0 518 262
        1 175 431
```

In [73]:
```
accuracy_svm <- mean(preds_svm == test$y_binary)
accuracy_svm
```

0.684704184704185

Model Performance Overview

In [74]:
```
model_results <- data.frame(
  Model = c("Logistic Regression", "Decision Tree", "Random Forest", "XGBoost",
  Accuracy = c(0.69, 0.70, 0.69, 0.65, 0.68),
  Precision = c(0.73, NA, 0.72, 0.66, NA),
  Recall = c(0.61, NA, 0.62, 0.64, NA),
  F1_Score = c(0.67, NA, 0.67, 0.65, NA)
)

model_results
```

A data.frame: 5 × 5

Model	Accuracy	Precision	Recall	F1_Score
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
Logistic Regression	0.69	0.73	0.61	0.67
Decision Tree	0.70	NA	NA	NA
Random Forest	0.69	0.72	0.62	0.67
XGBoost	0.65	0.66	0.64	0.65
SVM	0.68	NA	NA	NA

Model Performance Summary Table

Model	Accuracy	Precision	Recall	F1-Score	Notes
Logistic Regression	69%	73%	61%	0.67	Good precision, moderate recall
Decision Tree	70%	—	—	—	Slightly better recall than Logistic Regression
Random Forest	69%	72%	62%	0.67	Stable model, similar to Logistic Regression
XGBoost	65%	66%	64%	0.65	Balanced but lowest performance overall
SVM	68%	—	—	—	Comparable accuracy to RF/LR

Best Models

- **Best accuracy**: Decision Tree (~70%).
- **Best precision**: Logistic Regression (~73%).
- **Best recall**: XGBoost (~64%), though accuracy is lower.
- **Most balanced overall**: Logistic Regression and Random Forest (both ~69% accuracy, F1 ~0.67).

Conclusion

This project focused on developing a predictive model to determine whether clients will subscribe to a term deposit, utilizing a dataset comprising demographic, campaign-related, financial, and economic variables. Through thorough data preprocessing, exploratory analysis, and testing of various classification models (logistic regression, decision trees, random forests, XGBoost, SVM), the study identified several important findings:

- Customer Demographics and Financial Characteristics Influencing Subscription: Factors such as age, education level, and whether the customer has a housing or

personal loan significantly influence subscription likelihood. For instance, higher education levels correlated with increased subscription rates, aligning with the hypothesis that more educated clients are more receptive.

- Campaign-Related Variables Affecting Response: Variables such as the duration of the last contact and the number of contacts during the current campaign were strong predictors. Notably, longer contact durations and recent contacts increased the chances of subscription, emphasizing the importance of engagement quality.

- Economic and External Factors: Broader economic indicators like employment variation rate and consumer confidence index showed a subtle but relevant impact, suggesting external economic conditions influence customer responsiveness.

- Model Performance: Among the models tested, ensemble techniques like Random Forest and XGBoost achieved higher accuracy, precision, and recall scores, enabling more reliable prediction of client behavior. This demonstrates the feasibility of building effective models for targeted marketing.

Recommendations

1. Prioritize High-Probability Customers: Focus marketing efforts on clients with demographic and financial profiles associated with higher subscription rates, such as higher education and existing loans.
2. Enhance Contact Strategies: Allocate more resources to contacts with longer durations and recent contact history, as these signals are associated with higher success probabilities.
3. Leverage Predictive Models: Integrate the most accurate models (e.g., Random Forest) into the campaign decision process to optimize resource allocation and improve ROI.
4. Timing and Campaign Planning: Consider external economic indicators and optimal contact timings to increase engagement effectiveness.

Limitations

- Data Constraints: The dataset, while comprehensive, contains some missing or categorical variables that may limit the model's predictive power.
- Model Generalization: Models trained on historical data may not fully capture future market dynamics or external shocks.
- Contact Duration Variable: Duration, a strong predictor, is only known after the call, limiting its usefulness for real-time decision-making.
- External Factors: Economic indicators were only mildly impactful, suggesting external macroeconomic factors may require more granular data for accurate modeling.

Future Work

- Incorporate Additional Data:Integrate more recent economic indicators and customer engagement data for better model accuracy and robustness.

- Real-Time Prediction Deployment:Develop applications that provide real-time scoring during contact calls, considering variables available beforehand (excluding duration).
- Model Explainability: Focus on explainability techniques (e.g., SHAP values) to better understand feature influences and support transparent decision-making.
- Cross-Channel Analysis: Study how different communication channels or personalized messaging influence response rates.
- Longitudinal Studies: Analyze how customer behavior evolves over multiple campaigns to refine targeting strategies further.

In [ ]:

```

```